RelayMesh

PortalDot V3 routing and proof

Transfers that are proven, not reported.

RelayMesh signs the exact planned call, reads both chains to confirm the funds landed, and anchors a receipt anyone can re-hash.

Built by Saber PortalDot Hacker House 2026 Multi-Chain & Cross-Chain Infra

The problem

“Completed” is a claim, not a proof.

Status pages grade themselves

Wallets and bridges report “done” from their own records. Nobody independent checks the other side.

Two account worlds, one chain

On PortalDot V3, Substrate (5…) and EVM (0x…) accounts sit side by side. Moving value between them should be provable.

Disputes have no evidence

When funds go missing, users hold a screenshot and a support ticket, not a record anyone can verify.

Why PortalDot

V3 makes both sides checkable

One chain, two sides

Substrate and EVM accounts on the same chain, linked by Revive's account mapping. One tool can read both.

Same block, both views

A V3 block has the same number on both sides, so a transfer and its delivery are measured at the same finalized block.

Final means final

GRANDPA finality: once a block is final it cannot be rolled back, so a settled transfer stays settled.

Hash first, like Preimage

V3 governance registers a call's hash before running it. RelayMesh does the same for transfers: only the exact planned call counts.

How it works

Five checks between intent and receipt

01

Quote

The real V3 network fee, plus RelayMesh's 0.2 tPOT fee, stated up front.

02

Plan

Your browser decodes the exact call and checks amount, destination and fee before you sign.

03

Source

The server reads the block and accepts only the planned call, signed by you.

04

Delivery

After finality, the recipient's EVM balance is measured before and at that block.

05

Receipt

Append-only, and anchored on chain as soon as it settles, for about 0.02 tPOT.

Settled only when the amount that **actually arrived** matches. No provider tells RelayMesh what happened.

Proof, live on the V3 testnet

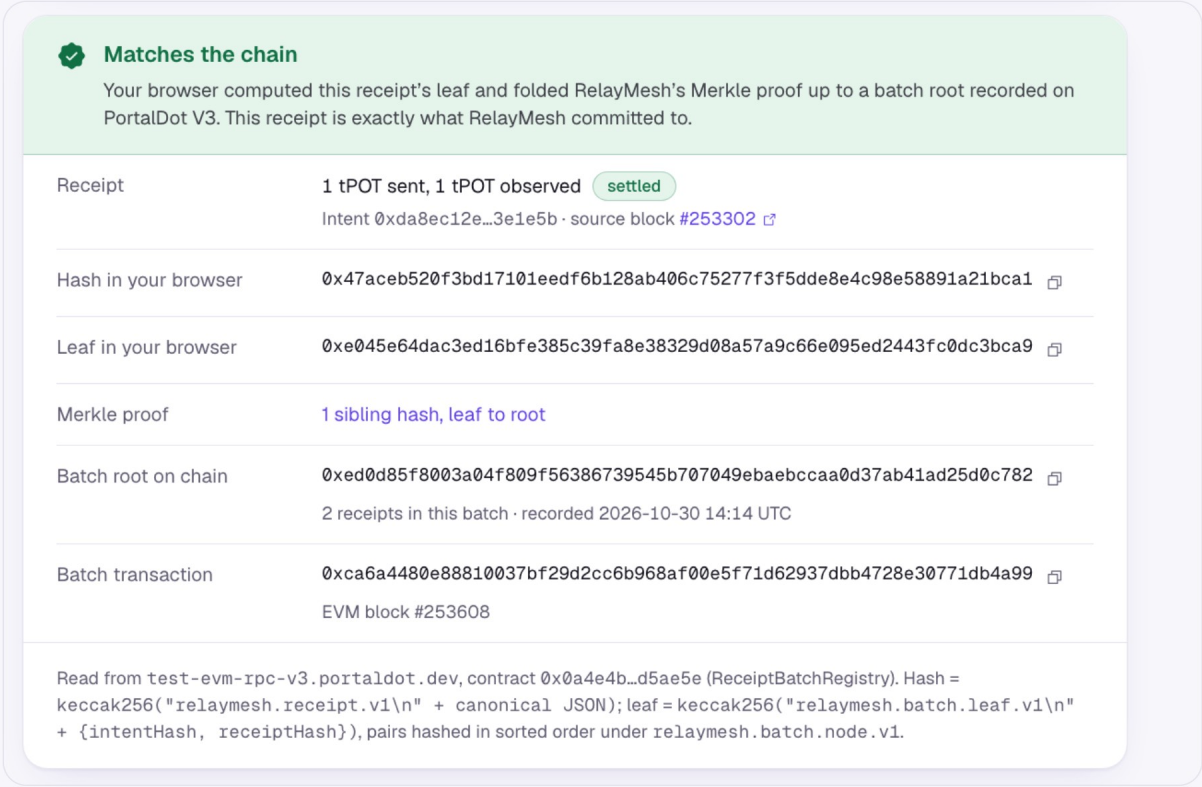
Real transfers, verified on both sides

9

signed transfers settled end to end, 372 tPOT in all. Zero mock data.

Anchored at	What it holds
IntentRegistry	First 5 receipts, one by one (1, 150, 8, 90, 40 tPOT)
Batch #244183	1 tPOT receipt, the first batch
Batch #252961	80 tPOT receipt and a bridge check
Batch #253608	2 receipts that paid the RelayMesh fee
Log #291548	A bridge check, the first 0.019 tPOT anchor

New anchors go to ReceiptBatchLog at 0x971d45f0…833b4d5 on PortalDot V3. Every transaction is on PortalDot's official explorer.



Receipts

Anyone can re-check one

1. Your browser hashes the receipt and computes its leaf.
2. It folds the Merkle proof up to a root.
3. It finds that root in the anchoring transaction on V3.

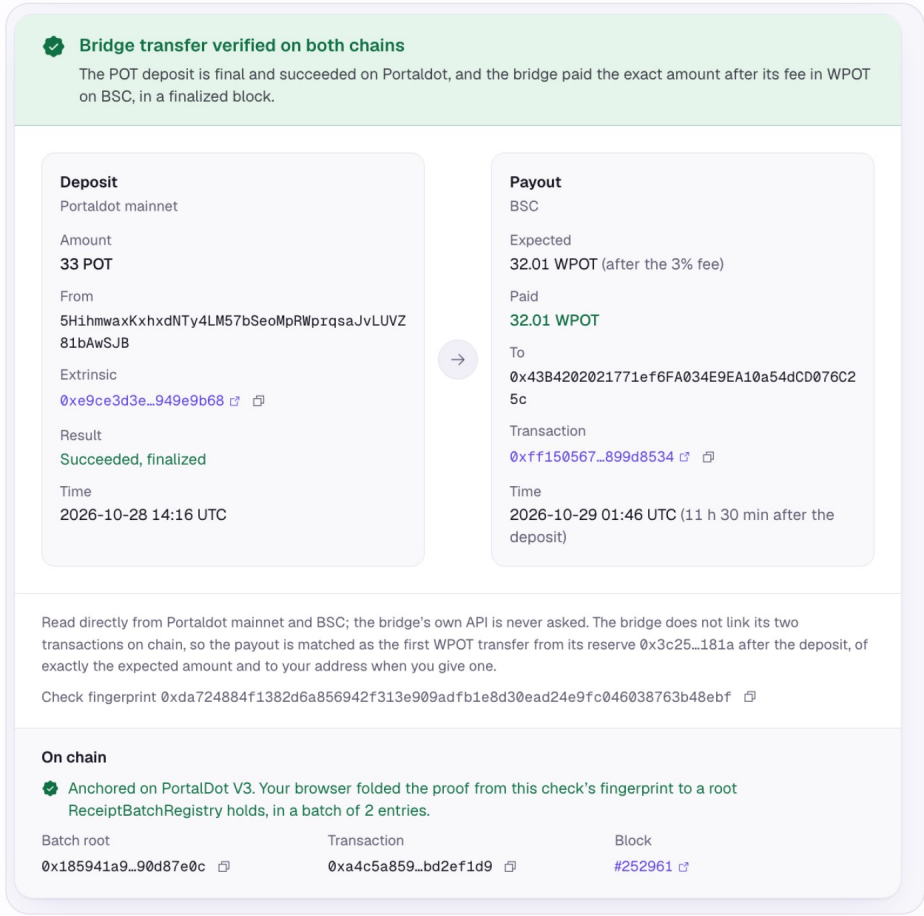
RelayMesh only supplies the proof, never the verdict. Change one digit and the root is one nobody recorded. Older receipts verify against IntentRegistry the same way.

Bridge checker · live on mainnet

Did the bridge pay out?

- Reads the POT deposit on Portaldot mainnet and the WPOT payout on BSC. The bridge's API is never asked.
- Exact after the 3% fee: 33 POT in, 32.01 WPOT out, 11 h 30 min later.
- Each verified check is anchored on V3, in the same batches as receipts.

PortalDot's public bridge has handled hundreds of transfers. Until now, nobody could prove one landed.



Sustainability · live since 30 October

Each transfer pays for its own proof

RelayMesh fee

0.2

tPOT per transfer, paid to the relayer

One anchor on chain

0.019

tPOT, measured on V3 (was 3.86)

One fee pays for

10

anchors, at any volume

Verified on V3

+0.40

tPOT reached the relayer from 2 real transfers

One signature: **utility.batchAll** of the transfer and the fee, so both happen or neither does. The recipient still gets the full amount, your browser checks the fee before you sign, and the fee is recorded inside the receipt that goes on chain. Each transfer is anchored as soon as it settles.

Robustness · 202 automated tests

Built for a real chain, not a happy path

Chain forks

A real fork hit block #172004. RelayMesh now finds the transfer in the block that won, instead of giving up.

Unreliable nodes

A public BSC node returned no logs 7 times in 12, so searches now merge parallel answers. A stuck V3 connection restarts itself.

Chain stalls

V3 stopped for 13 hours. A transfer signed during the stall stayed open and settled after blocks resumed.

Nothing guessed

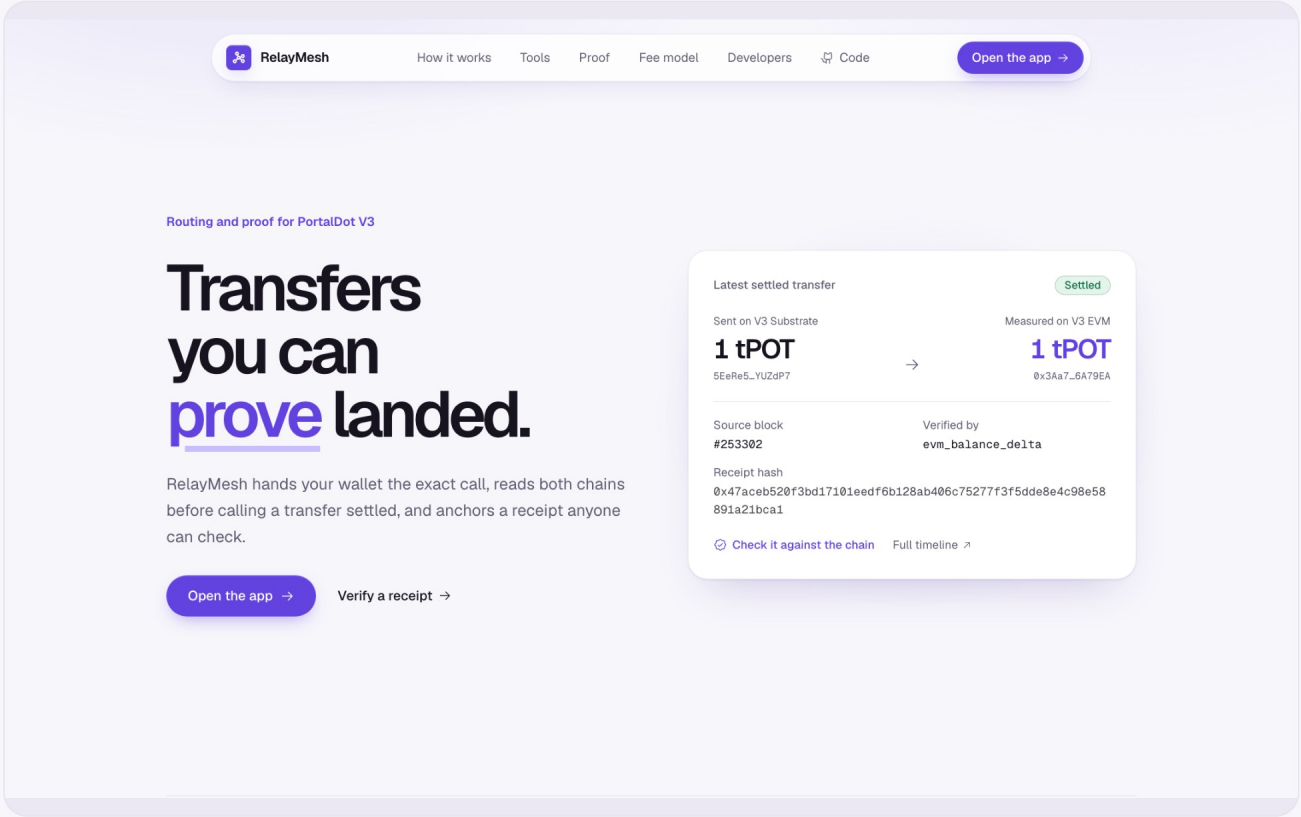
Anything RelayMesh cannot prove is held for review, with both observations kept in an append-only history.

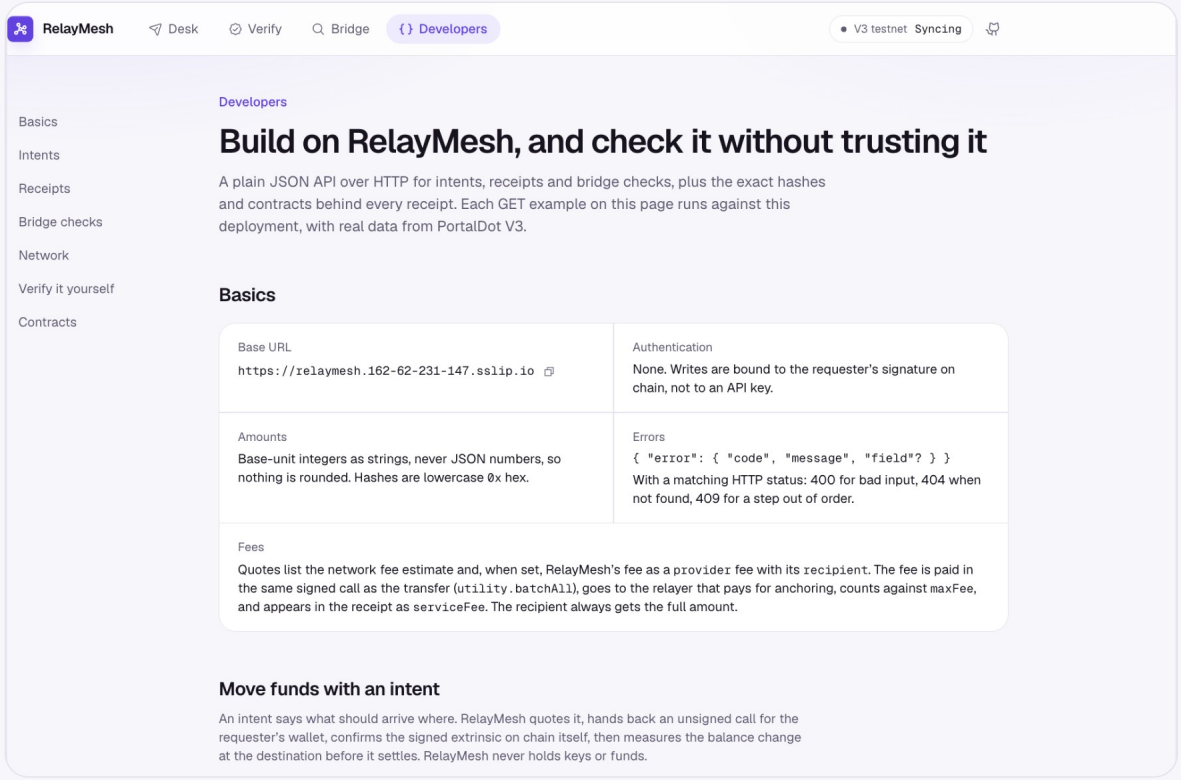
Live product

Running nonstop on V3

- Desk, Verify, Bridge and Developers
- Watcher settles, anchors in batches and restarts itself on a bad connection
- Landing page shows live figures and the fee ledger

`relaymesh.162-62-231-147.sslip.io`





For builders

An open API, checkable without us

- JSON API for intents, receipts, anchors and bridge checks
- Every read endpoint runs live on the page
- Hash rules and contracts published, plus a short viem script that verifies a receipt with no RelayMesh code

Ecosystem value

What PortalDot gets

Live today

A verification layer

Wallets, dApps and treasuries can prove a V3 transfer landed, without trusting any one provider.

Live today

Bridge transparency

Any POT to BSC transfer through PortalDot's bridge can be checked from both chains and anchored on V3. iBridge plugs in when it opens.

Planned

Proof for payouts

OpenGov treasury spends could carry a receipt proving the funds reached their recipient.

PortalDot confirmed iBridge stays closed to third parties through demo day, so bridge checks run against its public bridge on mainnet today.

Roadmap

Shipped, and what comes next

Shipped since Week 2

- Receipts anchored in Merkle batches
- Bridge checker, its checks anchored on V3
- A fee that funds anchoring, at 0.019 tPOT per anchor
- Developers page and open API

Next

- EVM to Substrate route, so quotes compete
- Resolve-review flow, recorded as evidence
- iBridge adapter as soon as it opens

After the Hacker House

- SDK and webhooks for wallets and dApps
- Mainnet deployment
- Receipts for treasury payouts

RelayMesh

Proven, not reported.

Live app
relaymesh.162-62-231-147.sslip.io

Code
github.com/Saber1Y/RelayMesh

Receipts on chain
0x971d45f0…833b4d5

Thank you. Built by Saber for the PortalDot Hacker House 2026.